BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Objective) Group - I

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 5477

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Rate of flow of heat is:

(A) Q/t^2
(B) Q^2/t
(C) $Q \times t$
(D) Q/t

حرارت کے بہاؤ کی شرح ہے:

Q2. In solids heat is transferred by:

(A) Convection
(B) Radiation
(C) Conduction
(D) Absorption

ٹھوس اجسام میں انتقال حرارت کا طریقہ ہے:

(A) کنویکشن
(B) ریڈی ایشن
(C) کنڈکشن
(D) ابزورپشن

Q3. Linear thermal expansion in solids (L) = :

(A) $T_o(1 + \alpha\Delta L)$ (B) $L_o(1 + \alpha\Delta T)$

(C) $T_o(\alpha\Delta L)$ (D) $L_o(1 - \alpha\Delta T)$

ٹھوس اجسام میں طولی پھیلاؤ (L):

Q4. Liquid pressure at depth h = :

(A) ρgh (B) ρgv

(C) ρgv^2 (D) mgh

h گہرائی پر مائع کا پریشر = :

Q5. Potential energy (P.E.) = :

(A) $1/2\ mv^2$ (B) mv^2

(C) m^2gh (D) mgh

پوٹینشل انرجی (P.E.):

Q6. 1 horse power is equal to:

(A) 674 W (B) 647 W

(C) 746 W (D) 764 W

ایک ہارس پاور برابر ہے:

Q7. Formula to find the orbital speed of a low orbit satellite is:

(A) $v_o = \sqrt{gR}$ (B) $v_o = gR$

(C) $v_o = \sqrt{gR^2}$ (D) $v_o = gR^2$

نچلے آربٹ میں گردش کرنے والے سیٹلائٹ کی آربٹل سپیڈ معلوم کرنے کا فارمولا ہے:

Q8. Conditions of equilibrium are:

(A) 1 (B) 2

(C) 3 (D) 4

ایکوی لبریم کی شرائط ہوتی ہیں:

Q9. Coefficient of friction (μ) is equal to:

(A) R/F_s (B) F_sR

(C) F_s/R (D) F_sR^2

فرکشن کا کوایفیشینٹ (μ) برابر ہے:

Q10. SI unit of momentum is:

(A) $kgms^{-1}$
(B) $kgm^{-1}s^{-1}$
(C) $kgms^{-2}$
(D) $kgms^{2}$

مومنٹم کا SI یونٹ ہے:

Q11. The product of velocity and time is equal to:

(A) Mass
(B) Displacement
(C) Acceleration
(D) Force

ولاسٹی اور وقت کا حاصل ضرب برابر ہے:

(A) ماس
(B) ڈسپلیسمنٹ
(C) ایکسلریشن
(D) فورس

Q12. Amount of substance is measured in:

(A) Gram
(B) Kilogram
(C) Newton
(D) Mole

مادے کی مقدار کی پیمائش کی جاتی ہے:

(A) گرام
(B) کلوگرام
(C) نیوٹن
(D) مول

Physics (Subjective) Group - I

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 5477

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) Define base and derived quantities.

(ii) What is meant by Vernier constant?

(iii) What is the difference between rest and motion?

(iv) Define acceleration and write its formula.

(v) Write two characteristics of physical quantities.

(vi) What is the law of inertia?

(vii) Define force and write its SI unit.

(viii) How can vector quantities be represented graphically?

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Define torque. Write its mathematical formula.

(ii) State and explain Principle of Moments.

(iii) Define rigid body and axis of rotation.

(iv) What is gravitational constant? Write its value in SI unit.

(v) On what factors orbital speed of satellite depends?

(vi) Define efficiency. Also write its formula.

(vii) What is meant by the term power? Define 1 watt.

(viii) What is meant by density? Write its formula.

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) Define pressure and write its SI unit.

(ii) Differentiate between stress and strain.

(iii) What is Hooke's law? What is meant by elastic limit?

(iv) Define the terms heat and temperature.

(v) Define latent heat of vaporization.

(vi) Why conduction of heat does not take place in gases?

(vii) How does heat reach us from the Sun?

(viii) Write two uses of convection currents.

> Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

> حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 04) Derive the second equation of motion with the help of speed-time graph:

Q5(b). (Marks 05) A cyclist of mass 40kg exerts a force of 200N to move his bicycle with an acceleration of $3ms^{-2}$. How much is the force of friction between the road and the tyres?

Q6(a). (Marks 04) Define kinetic energy and derive its equation.

Q6(b). (Marks 05) A picture frame is hanging by two vertical strings. The tensions in the strings are 3.8N and 4.4N. Find the weight of the picture frame.

Q7(a). (Marks 04) Explain the pressure in liquids and derive its formula ($P = \rho gh$).

Q7(b). (Marks 05) How much ice will melt by 50000J of heat? Latent heat of fusion of ice = $336000 Jkg^{-1}$.

GROUP – II

Physics (Objective) Group - II

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** 5472

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The amount of a substance in terms of numbers is measured:

(A) Gram
(B) Kilogram
(C) Newton
(D) Mole

کسی شے میں مادے کی مقدار معلوم کرنے کا یونٹ ہے:

(A) گرام
(B) کلوگرام
(C) نیوٹن
(D) مول

Q2. A train is moving at a speed of $36 kmh^{-1}$. Its speed expressed in ms^{-1}:

(A) $10 ms^{-1}$
(B) $20 ms^{-1}$
(C) $25 ms^{-1}$
(D) $30 ms^{-1}$

ایک ٹرین $36 kmh^{-1}$ کی سپیڈ سے حرکت کر رہی ہے۔ ms^{-1} میں اس کی سپیڈ ہوگی:

Q3. The unit of momentum is:

(A) Nm
(B) $kgms^{-2}$
(C) Ns
(D) Ns^{-1}

مومنٹم کا یونٹ ہے:

Q4. A force of 10N is making an angle of 30° with x-axis. Its horizontal component will be:

(A) 4N
(B) 5N
(C) 7N
(D) 8.7N

10 نیوٹن کی ایک فورس x-ایکسز کے ساتھ 30° کا زاویہ بناتی ہے۔ اس کا افقی کمپوننٹ ہوگا:

Q5. The value of g on moon's surface is $1.6ms^{-2}$. What will be the weight of a 100kg body on the surface of moon?

(A) 100N
(B) 160N
(C) 1000N
(D) 1600N

چاند کی سطح پر g کی قیمت $1.6ms^{-2}$ ہے۔ چاند پر 100kg کے ایک جسم کا وزن ہوگا:

Q6. The K.E. of a body of mass 2kg is 25J. Its speed is:

(A) $5ms^{-1}$
(B) $12.5ms^{-1}$
(C) $25ms^{-1}$
(D) $50ms^{-1}$

2kg کے ایک جسم کی K.E. 25 جول ہے۔ اس کی سپیڈ ہوگی:

Q7. SI unit of pressure is Pascal which is equal to:

(A) 10^4Nm^{-2}
(B) $1Nm^{-2}$
(C) 10^2Nm^{-2}
(D) 10^3Nm^{-2}

پریشر کا SI یونٹ پاسکل ہے اور ایک پاسکل برابر ہوتا ہے:

Q8. Which affects evaporation?

(A) Temperature
(B) Surface area of liquid
(C) Wind
(D) All these

کون سا ایواپوریشن کو متاثر کرتا ہے؟

(A) درجہ حرارت
(B) مائع کی سطح کا رقبہ
(C) ہوا
(D) یہ تمام

Q9. The thermal conductivity of copper in $Wm^{-1}K^{-1}$:

(A) 200
(B) 300
(C) 400
(D) 430

کاپر کی $Wm^{-1}K^{-1}$ میں تھرمل کنڈکٹیویٹی ہے:

Q10. The value of coefficient of friction between tyre and dry road is: _____

(A) 0.6 (B) 0.8

(C) 0.2 (D) 1

ٹائر اور خشک سڑک کے درمیان کوایفیشینٹ آف فرکشن کی قیمت ہے: _____

Q11. The derived unit is:

(A) Ampere (B) Pascal

(C) Candela (D) Kelvin

ماخوذیونٹ ہے:

(A) ایمپیئر **(B)** پاسکل

(C) کینڈیلا **(D)** کیلون

Q12. The specific heat of sand is:

(A) $835 Jkg^{-1}K^{-1}$ (B) $2100 Jkg^{-1}K^{-1}$

(C) $4200 Jkg^{-1}K^{-1}$ (D) $900 Jkg^{-1}K^{-1}$

ریت کی حرارت مخصوصہ ہے:

Physics (Subjective) Group - II

Time Allowed: 01:45 Hours | **Maximum Marks:** 48 | **Paper Code:** 5472

Note: Part - I (حصہ اول)

Q2. (Marks 10) Write short answers to any FIVE parts.

(i) Differentiate between base and derived quantities.

(ii) Define prefixes and give one example.

(iii) Your age is 14 years. Convert your age in seconds.

(iv) What is meant by position?

(v) What is the difference between distance and displacement?

(vi) Define gravitational acceleration and write its value.

(vii) Define momentum and write its unit.

(viii) What is meant by coefficient of friction? Also write its symbol.

Q3. (Marks 10) Write short answers to any FIVE parts.

(i) Differentiate between line of action of force and moment arm.

(ii) What is meant by moment of force? Also write is it vector or scalar quantity?

(iii) Give an example of a body which is at rest but not in equilibrium.

(iv) Define geostationary orbit.

(v) Write two uses of artificial satellites.

(vi) Differentiate between artificial and natural satellites.

(vii) Define energy and write its unit.

(viii) Write the SI unit of power and also define the unit of power.

Q4. (Marks 10) Write short answers to any FIVE parts.

(i) What is meant by elasticity?

(ii) Define the term pressure.

(iii) What is tensile strain? Give its formula.

(iv) Define heat capacity.

(v) Define the terms heat and temperature.

(vi) Define convection.

(vii) Define thermal conductivity.

(viii) Define land breeze and sea breeze.

> Note: Part - II, Attempt any TWO questions. Each question carries 09 marks.

> حصہ دوم، کوئی سے دو سوالات کے جوابات تحریر کیجیے۔ ہر سوال کے 09 نمبر ہیں۔

Q5(a). (Marks 04) With the help of speed-time graph, prove that: $v_i = v_i + at$

Q5(b). (Marks 05) How much centripetal force is needed to make a body of mass 0.5kg to move in a circle of radius 50cm with a speed of $3ms^{-1}$?

Q6(a). (Marks 04) State and derive law of gravitation.

Q6(b). (Marks 05) Find the perpendicular components of a force of 50N making an angle of 30° with x-axis.

Q7(a). (Marks 04) Define and explain linear thermal expansion in solids. Also derive $L = L_o(1 + \alpha\Delta T)$ equation.

Q7(b). (Marks 05) A steel wire 1m long and cross-sectional area $5\times10^{-5}m^2$ is stretched through 1mm by a force of 10000N. Find the Young's modulus of the wire.